

DE-2024 — PRACTICE PAPER

Total Marks: 200 | Time: 150 Minutes | Question Booklet Code: B

100 Questions | +2 Correct | -0.50 Wrong / Multi-mark | 0 Not Attempted (E)

SECTION – 1

Physics · Chemistry · Computer Science · Environmental Science

001. What is the SI unit of Force?

- (A) Joule (B) Newton (C) Watt (D) Pascal

002. A vernier caliper's least count is 0.02 mm. Main scale reads 13 mm; vernier reads 4 div. Total reading = ?

- (A) 13.08 mm (B) 13.02 mm (C) 13.04 mm (D) 13.4 mm

003. 1 Newton = _____ dyne

- (A) 10^3 (B) 10^4 (C) 10^5 (D) 10^6

004. What is the CGS unit of Force?

- (A) Newton (B) Joule (C) Dyne (D) Erg

005. Five resistance readings (Ω): 2.10, 2.12, 2.08, 2.11, 2.09. Find mean absolute error.

- (A) 0.012 (B) 0.12 (C) 0.0012 (D) 1.2

006. Which instrument measures the outer diameter of a wire?

- (A) Vernier caliper (B) Micrometer screw gauge (C) Spherometer (D) Physical balance

007. Newton's Third Law states _____

- (A) $F = ma$ (B) Every action has equal & opposite reaction (C) $P = mv$ (D) $F = mv$

008. A car of mass 1000 kg accelerates at 2 m/s^2 . Net force applied = ?

- (A) 500 N (B) 2000 N (C) 200 N (D) 2 N

009. Which is an example of conservation of momentum?

- (A) Rocket propulsion (B) Free fall (C) Pendulum swing (D) Rolling friction

010. A body in horizontal circular motion experiences which inward force?

- (A) Centrifugal (B) Gravitational (C) Centripetal (D) Frictional

011. Which of the following is a contact force?

- (A) Gravitational (B) Magnetic (C) Friction (D) Electrical

012. Work done by a conservative force equals the change in _____

- (A) Kinetic energy (B) Total energy (C) Potential energy (D) Thermal energy

013. A 9 V battery drives 3 mA through resistor R. Find R.

- (A) $3 \text{ k}\Omega$ (B) $27 \text{ k}\Omega$ (C) $0.3 \text{ k}\Omega$ (D) 300Ω

014. Unit of electric field intensity is _____

- (A) N/C (B) C/N (C) N·C (D) J/C

015. Gauss's Law relates electric flux to _____

- (A) Charge enclosed (B) Distance (C) Area only (D) Permeability

016. Reciprocal of resistivity is known as _____

- (A) Resistance (B) Conductance (C) Conductivity (D) Capacitance

017. 6Ω and 12Ω in parallel — equivalent resistance = ?

- (A) 18Ω (B) 6Ω (C) 4Ω (D) 2Ω

018. A capacitor stores energy in _____

- (A) Electric field (B) Magnetic field (C) Gravitational field (D) None of these

019. Arrange in decreasing order of electrical conductivity: Gold, Silver, Copper, Aluminium

- (A) Silver, Copper, Gold, Al (B) Gold, Silver, Copper, Al (C) Copper, Silver, Gold, Al (D) Al, Copper, Silver, Gold

020. Normal body temperature is 98.6°F. Convert to Celsius.

- (A) 36°C (B) 37°C (C) 38°C (D) 40°C

021. Which substance has the highest specific heat capacity?

- (A) Iron (B) Copper (C) Water (D) Aluminium

022. When temperature decreases, resistance of a metallic conductor _____

- (A) Increases (B) Decreases (C) Remains constant (D) Becomes zero

023. Latent heat formula is _____

- (A) $Q = mL$ (B) $Q = mcT$ (C) $Q = c/mT$ (D) $L = Q/T$

024. Convert 300 K to Celsius.

- (A) 27°C (B) 37°C (C) -27°C (D) 573°C

025. Which of the following CAN travel through vacuum?

- (A) Sound waves (B) Seismic waves (C) Electromagnetic waves (D) None of above

026. Radio waves have _____ wavelength compared to X-rays.

- (A) Shorter (B) Same (C) Longer (D) No relation

027. Frequency 500 Hz, wavelength 0.68 m. Speed of the wave = ?

- (A) 340 m/s (B) 34 m/s (C) 3400 m/s (D) 0.34 m/s

028. LASER stands for _____

- (A) Light Amplification by Stimulated Emission of Radiation (B) Light Absorption by Simulated Electron Radiation (C) Large Amplitude Stimulated Electromagnetic Radiation (D) Light Amplification by Stimulated Electronic Reaction

029. Which phenomenon causes the blue colour of the sky?

- (A) Refraction (B) Diffraction (C) Scattering (D) Total internal reflection

030. Sabine's formula for reverberation time is $T =$ _____

- (A) $0.165 V / A$ (B) $0.165 A / V$ (C) $V / 0.165A$ (D) $A / 0.165V$

031. Electrolysis of water produces _____

- (A) H_2 and O_2 (B) H_2 and N_2 (C) O_2 and CO_2 (D) Only H_2

032. $CaCO_3 \rightarrow CaO + CO_2$ is which type of reaction?

- (A) Combination (B) Displacement (C) Decomposition (D) Double displacement

033. Balanced equation for $Fe + HCl \rightarrow FeCl_2 + H_2$ is _____

- (A) $Fe + 2HCl \rightarrow FeCl_2 + H_2$ (B) $2Fe + HCl \rightarrow FeCl_2 + H_2$ (C) $Fe + HCl \rightarrow FeCl + H_2$ (D) $Fe + 2HCl \rightarrow 2FeCl + H_2$

034. Bones and teeth are mainly made up of _____

- (A) Calcium carbonate (B) Calcium phosphate (C) Potassium nitrate (D) Sodium chloride

035. pH of a neutral solution is _____

- (A) 0 (B) 14 (C) 7 (D) 5

036. Non-metal oxide + water $\rightarrow$ _____

- (A) Salt (B) Base (C) Acid (D) Neutral solution

037. Molecular formula of Sodium Sulphate is _____

- (A) $NaSO_4$ (B) Na_2SO_4 (C) Na_2SO_3 (D) $NaSO_3$

038. Electrons in the outermost shell of Chlorine = ?

- (A) 7 (B) 8 (C) 2 (D) 6
039. Coating iron with zinc to prevent rusting is called _____
 (A) Painting (B) Galvanization (C) Greasing (D) Anodizing
040. Which metal is the best conductor of electricity?
 (A) Copper (B) Gold (C) Silver (D) Aluminium
041. Which is NOT a valid data type in MS-Excel?
 (A) Number (B) Date/Time (C) Paragraph (D) Text
042. Default number of slides in a new MS-PowerPoint file is _____
 (A) 0 (B) 1 (C) 2 (D) 5
043. Keyboard shortcut to Save in MS-Word is _____
 (A) Ctrl+P (B) Ctrl+S (C) Ctrl+Z (D) Ctrl+X
044. RAM stands for _____
 (A) Read Access Memory (B) Random Access Memory (C) Rapid Access Module (D) Read And Modify
045. Which of the following is an input device?
 (A) Monitor (B) Printer (C) Speaker (D) Keyboard
046. Acid rain is mainly caused by _____
 (A) CO₂ and O₂ (B) SO₂ and NO₂ (C) CH₄ and N₂ (D) O₂ and H₂
047. The ozone layer is found in _____
 (A) Troposphere (B) Mesosphere (C) Thermosphere (D) Stratosphere
048. Greenhouse effect is mainly caused by increase in _____
 (A) Oxygen (B) Carbon dioxide (C) Nitrogen (D) Hydrogen
049. Which is a NON-renewable source of energy?
 (A) Solar (B) Wind (C) Coal (D) Tidal
050. Which pollutant is mainly responsible for photochemical smog?
 (A) Carbon monoxide (B) Sulphur dioxide (C) Nitrogen oxides (D) Methane

SECTION – 2

Mathematics · English Communication · Grammar

051. If $A = \begin{bmatrix} 3 & 2 \\ 1 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 0 \\ 2 & 3 \end{bmatrix}$, then $A + B =$ _____
 (A) $\begin{bmatrix} 4 & 2 \\ 3 & 7 \end{bmatrix}$ (B) $\begin{bmatrix} 4 & 3 \\ 3 & 7 \end{bmatrix}$ (C) $\begin{bmatrix} 3 & 2 \\ 4 & 7 \end{bmatrix}$ (D) $\begin{bmatrix} 4 & 2 \\ 4 & 7 \end{bmatrix}$
052. If $A = \begin{bmatrix} 2 & 1 \\ 5 & 3 \end{bmatrix}$, then $|A| =$ _____
 (A) 1 (B) -1 (C) 11 (D) 6
053. If $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$, then $A^{-1} =$ _____
 (A) $(1/-2)\begin{bmatrix} 4 & -2 \\ -3 & 1 \end{bmatrix}$ (B) $(1/2)\begin{bmatrix} 4 & 2 \\ 3 & 1 \end{bmatrix}$ (C) $(1/-2)\begin{bmatrix} 4 & 2 \\ 3 & 1 \end{bmatrix}$ (D) $\begin{bmatrix} -2 & 1 \\ 1.5 & -0.5 \end{bmatrix}$
054. If $A = \begin{bmatrix} 1 & 0 \\ 2 & 3 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & 1 \\ 0 & 4 \end{bmatrix}$, find $2A - B$.
 (A) $\begin{bmatrix} 0 & -1 \\ 4 & 2 \end{bmatrix}$ (B) $\begin{bmatrix} 0 & 1 \\ 4 & 2 \end{bmatrix}$ (C) $\begin{bmatrix} 0 & -1 \\ 4 & -2 \end{bmatrix}$ (D) $\begin{bmatrix} 2 & 0 \\ 4 & 2 \end{bmatrix}$
055. $\cos(300^\circ) =$ _____
 (A) $1/2$ (B) $-1/2$ (C) $\sqrt{3}/2$ (D) $-\sqrt{3}/2$
056. If $\sin\theta = 5/13$, θ in first quadrant, then $\tan\theta =$ _____
 (A) $5/12$ (B) $12/13$ (C) $5/13$ (D) $13/12$
057. $\cos^{-1}(-1/2) =$ _____
 (A) $\pi/3$ (B) $2\pi/3$ (C) $\pi/6$ (D) $5\pi/6$

058. If $x + y = 5$ and $x - y = 3$, find $|x||y|\cos\theta$ (dot product relation).

- (A) $\pi/3$ (B) $\pi/4$ (C) $\pi/6$ (D) $\pi/2$

059. Vectors $(1, 2, a)$ and $(2, -1, 3)$ are perpendicular. Find a .

- (A) 0 (B) $4/3$ (C) $-4/3$ (D) 1

060. Centre of circle $x^2 + y^2 + 4x - 6y + k = 0$ is $(-2, 3)$. Find k .

- (A) 0 (B) 13 (C) -13 (D) 5

061. Equation of line through $(2,3)$ and $(4,7)$ is _____

- (A) $y = 2x - 1$ (B) $y = 2x + 1$ (C) $y = x + 1$ (D) $2y = x + 1$

062. $\lim_{x \rightarrow 0} \sin 3x / x =$ _____

- (A) 1 (B) 3 (C) 0 (D) ∞

063. $\lim_{n \rightarrow \infty} (3n^2 + 5) / (6n^2 + 1) =$ _____

- (A) $1/2$ (B) $3/6$ (C) 5 (D) 0

064. $f(x) = \log_{10}(x)$. Find $f(81)$.

- (A) 3 (B) 4 (C) 27 (D) $1/4$

065. $d/dx (\sin^2 x) =$ _____

- (A) $2\sin x$ (B) $\sin 2x$ (C) $2\cos x$ (D) $\cos 2x$

066. $d/dx [\log(x^2 + 4)] =$ _____

- (A) $2x/(x^2+4)$ (B) $1/(x^2+4)$ (C) $x/(x^2+4)$ (D) $2/(x+4)$

067. If $x = \sin\theta$ and $y = \cos\theta$, then $dy/dx =$ _____

- (A) $-\cot\theta$ (B) $\tan\theta$ (C) $\cot\theta$ (D) $-\tan\theta$

068. Condition for $f(x)$ to have a maximum at $x = a$ is _____

- (A) $f'(a)=0$ and $f''(a)>0$ (B) $f'(a)=0$ and $f''(a)<0$ (C) $f'(a)>0$ (D) $f'(a)<0$

069. $\int \sin(5x+3) dx =$ _____ + C

- (A) $-(1/5)\cos(5x+3)$ (B) $(1/5)\cos(5x+3)$ (C) $-5\cos(5x+3)$ (D) $5\cos(5x+3)$

070. $\int_0^{\pi/2} \cos x dx =$ _____

- (A) 0 (B) 1 (C) -1 (D) $\pi/2$

071. $\int e^x (\sin x + \cos x) dx =$ _____ + C

- (A) $e^x \sin x$ (B) $e^x \cos x$ (C) $e^x (\sin x - \cos x)$ (D) $2e^x$

072. $\int (2x+3)/(x^2+3x+5) dx =$ _____ + C

- (A) $\log|x^2+3x+5|$ (B) $(1/2)\log|x^2+3x+5|$ (C) $2\log|x^2+3x+5|$ (D) $\log(2x+3)$

073. $\log_{10}(625) =$ _____

- (A) 4 (B) 5 (C) 3 (D) 25

074. $\log_{10} 8 + \log_{10} 9 =$ _____

- (A) 5 (B) 6 (C) 7 (D) 4

075. Mean of observations 19, 15, 12, k , 8, 3 is 11. Find k .

- (A) 8 (B) 9 (C) 10 (D) 11

Read the following passage carefully and answer Q.No. 076 to 080:

The Internet has transformed the way people communicate, learn, and do business. It began as a small research network called ARPANET in the late 1960s in the United States, primarily used by the military and academic institutions. Over the decades, it evolved into a global network connecting billions of devices. The World Wide Web (WWW), invented by British scientist Tim Berners-Lee in 1989, made the Internet accessible to the general public. Today, the Internet supports e-commerce, social media, cloud computing, and online education. While it has brought enormous benefits, it has also raised concerns about privacy,

cybercrime, and digital addiction.

076. ARPANET was originally used by _____

- (A) General public (B) Military and academic institutions (C) Commercial businesses (D) Schools only

077. The World Wide Web was invented by _____

- (A) Bill Gates (B) Mark Zuckerberg (C) Tim Berners-Lee (D) Steve Jobs

078. In which year was the World Wide Web invented?

- (A) 1969 (B) 1989 (C) 1999 (D) 1979

079. Which of the following is a concern raised by the Internet?

- (A) Better communication (B) Online education (C) Cybercrime (D) E-commerce

080. ARPANET was established in which country?

- (A) United Kingdom (B) China (C) United States (D) Germany

081. Verbal communication can be in _____ form.

- (A) Written (B) Oral and written (C) Oral (D) None of these

082. Pitch, tone, speed of voice are part of _____

- (A) Body language (B) Sign language (C) Paralanguage (D) Code language

083. Communication is a _____ process.

- (A) Two-way (B) Three-way (C) One-way (D) Single-way

084. Hindrance in communication at every stage is known as _____

- (A) Barrier (B) Noise (C) Sender (D) Receiver

085. Our physical appearance is an example of _____ communication.

- (A) Oral (B) Written (C) Verbal (D) Non-verbal

086. What is written at the top-left of a business letter?

- (A) Sender's address (B) Receiver's address (C) Date (D) Salutation

087. Where is the salutation placed in a business letter?

- (A) Just above the date (B) Just below the date (C) Along with subject (D) None of these

088. Which is NOT a part of a business letter?

- (A) Photo (B) Salutation (C) Date (D) Signature

089. A letter is incomplete and worthless without _____

- (A) Signature (B) Postscript (C) Enclosure (D) Photo

090. Full form of C.O.D. is _____

- (A) Cash on Delay (B) Cash on Delivery (C) Cash on Order (D) Cash on Distance

091. Oh! What a beautiful painting. Identify the interjection.

- (A) Painting (B) Beautiful (C) Oh! (D) The whole sentence

092. The cat is sitting _____ the chair. (Select appropriate option)

- (A) between (B) under (C) into (D) up

093. She _____ to work every day. (Select appropriate option)

- (A) go (B) goes (C) gone (D) going

094. The jury _____ divided in its opinion. (Select appropriate option)

- (A) are (B) is (C) has (D) have

095. Do you have _____ questions? (Select appropriate option)

(A) some

(B) any

(C) many

(D) much

096. Select the correct spelling:

(A) Grammer

(B) Gramer

(C) Grammar

(D) Grammir

097. Select the correct spelling:

(A) Accomodation

(B) Accommodation

(C) Acomodation

(D) Acommmodation

098. Choose the correct grammatical sentence:

(A) He don't know the answer.

(B) He doesn't knows the answer.

(C) He doesn't know the answer.

(D) He not know the answer.

099. Choose the correct grammatical sentence:

(A) Neither he nor she are correct.

(B) Neither he nor she is correct.

(C) Neither he nor she was correct.

(D) Neither he or she is correct.

100. Choose the correct grammatical sentence:

(A) Each of the students have submitted their work.

(B) Each of the students has submitted their work.

(C) Each of the students have submitted his work.

(D) Each of the students has submitted there work.

◆ SPACE FOR ROUGH WORK ◆

— END OF PRACTICE PAPER —